Hansol Suh

Last updated: September 14, 2026

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Education

Georgia Institute of Technology Atlanta, GA

MSc, Computational Science and Engineering Fall 2022

PhD studies, Computational Science and Engineering: Spring 2018–Fall 2022. Program concluded with an MSc. Advisor: Tobin Isaac. Research: numerical optimization and HPC.

BSc, Civil Engineering, Highest Honor August 2011–July 2016

Technical Skills

Programming: C, C++, Python, Julia, Java, MATLAB; CUDA and HIP.

Methods and tools: MPI, OpenMP, Unix/Linux, Git, machine learning, parallel computing.

Publications

Peer-reviewed publications

Verification of PETSc with CIVL Using LLM-Generated ACSL Contracts and Deterministic Driver Generation

H. Suh, J. Hückelheim, and S. Siegel.

Correctness 2026: 10th International Workshop on Software Correctness for HPC Applications (SC26). Accepted; workshop scheduled for November 16, 2026.

arXiv:2609.31687

PETSc/TAO Developments for GPU-Based Early Exascale Systems

R. T. Mills, M. F. Adams, S. Balay, J. Brown, J. Faibussowitsch, T. Isaac, M. G. Knepley, T. Munson, **H. Suh**, S. Zampini, H. Zhang, and J. Zhang.

The International Journal of High Performance Computing Applications. 2025.

doi:10.1177/10943420241303710

Evaluation of a Minimally Synchronous Algorithm for 2:1 Octree Balance

H. Suh and T. Isaac.

SC20: International Conference for High Performance Computing, Networking, Storage and Analysis. November 2020.

Posters

Discrete Adjoints in PFLOTRAN for Physics-Constrained Machine Learning

H. Suh et al.

AGU Annual Meeting 2026, San Francisco, CA. Submitted.

Preprints and manuscripts

A Portable and Versatile Limited-Memory BFGS Implementation in PETSc/TAO

H. Suh, T. Isaac, A. Dener, T. Munson, H. Zhang, and R. T. Mills.

arXiv:2608.00196. 2026. Submitted to *Computational Optimization and Applications*.

Variable Metric Three Operator Splitting

H. Suh and T. Isaac.

arXiv:2610.38935. 2026.

A Heuristic for Parameter Selection in Adaptive Relaxed ADMM Based on Numerical Ranges

T. Isaac and **H. Suh**.

Manuscript.

Technical Reports and Manuals

PETSc/TAO Users Manual, Revision 3.25

S. Balay et al. (including **H. Suh**).

ANL-21/39 Rev. 3.25. Argonne National Laboratory. 2026.

TAO Users Manual, Revision 3.15

A. Dener, A. Denchfield, **H. Suh**, T. Munson, J. Sarich, S. Wild, S. Benson, and L. C. McInnes.

ANL/MCS-TM-322 Rev. 3.15. Argonne National Laboratory. March 2021.

Experience

Argonne National Laboratory Lemont, IL

Predoctoral Appointee March 2023–Present

- Develop dense BFGS implementations for GPUs in PETSc/TAO.
- Develop proximal optimization solvers for PETSc/TAO.
- Supervisors: Tobin Isaac (former); Richard Mills (current).

Summer Research Appointments 2019, 2021, 2022

- Research Aide, Technical (June–August 2022): developed Anderson-accelerated three-operator splitting and hierarchical, composable PETSc/TAO solvers. Supervisor: Todd Munson.
- Research Aide (May–August 2021): implemented variable-metric three-operator splitting in PETSc/TAO and evaluated the implementation in an HPC setting. Supervisor: Alp Dener.
- Givens Associate (May–August 2019): implemented ADMM and a variant in PETSc/TAO and evaluated their performance in an HPC setting. Supervisor: Alp Dener.

Georgia Institute of Technology Atlanta, GA

Graduate Research Assistant May 2018–December 2022

Research with Tobin Isaac in PETSc development, numerical optimization and analysis, p4est, and HPC.

Graduate Student August 2016–May 2018

Republic of Korea Army Icheon, South Korea

Anti-Aircraft Gunner; Armory Department. November 2013–August 2015

Grants & Funding

CS2: Modular Verification of Scientific Software May 2025–May 2029

Using Model Checking, Symbolic Execution, and Abstraction

Senior Personnel, NSF/DOE Correctness for Scientific Computing Systems (CS2), NSF 24-571. Award: $3,200,000. PIs: Stephen Siegel (University of Delaware), Jan Hückelheim (Argonne National Laboratory).

AI4Soil Summer 2026

A Deployable Artificial Intelligence Soil Prototype for Improving Land–Atmosphere Prediction. ANL LDRD Seed project; collaborator with Joshua Minai. Award: $25,000.

Projects

Terrestrial Wetland Function and Resilience Science Focus Area Summer 2025–Present

Developing adjoint-based sensitivity-analysis capabilities in PFLOTRAN for subsurface reactive transport, and physics-constrained machine-learning methods for wetland systems.

TaoTerm

PIONEER thrust: Computationally Efficient Composable Solvers for PETSc/TAO. Developed composable objects for the TAO optimizer, enabling flexible, on-the-fly modification of optimization objectives.

Presentations

TaoTerm: Composable Objective Function Terms

H. Suh.

Consortium for the Advancement of Scientific Software BoF Days, online. February 11, 2026.

Slides

Dense Broyden–Fletcher–Goldfarb–Shanno (BFGS)

H. Suh.

PETSc User Meeting 2025, Buffalo, NY. May 20, 2025.

Slides

An Introduction to PETSc/TAO

H. Suh.

ALCF Hands-On HPC Workshop, Lemont, IL. October 31, 2024.

Slides

Mini Tutorial: TaoADMM

H. Suh.

PETSc User Meeting 2023, Chicago, IL. June 7, 2023.

Slides

An Asynchronous Algorithm for 2:1 Octree Balance

H. Suh and T. Isaac.

SIAM Conference on Parallel Processing 2020, Seattle, WA. February 14, 2020.

Using the PETSc/TAO ADMM Methods on GPUs

H. Suh, A. Dener, T. Isaac, and T. Munson.

SIAM Conference on Parallel Processing 2020, Seattle, WA. February 13, 2020.

Teaching

Computational Data Analysis (ISYE 6740) Fall 2021, Fall 2022

Teaching Assistant.

Introduction to High Performance Parallel Computing (CX 4220) Spring 2021

Teaching Assistant.

High Performance Parallel Computing (CSE 6230) Fall 2019

Teaching Assistant.